

MEGA

MAGIC ELEMENTS IN GRAPHICAL ARTS

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Graphics Programming(2026) Final Team Project

PROJECT OVERVIEW

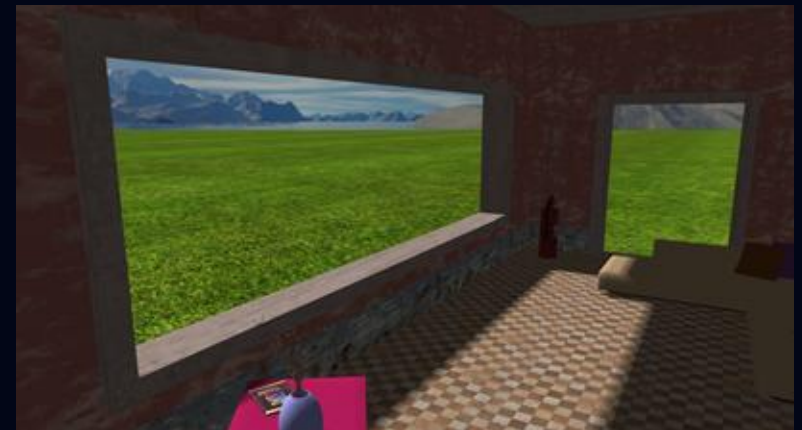
THE CORE CONCEPT

The project features a **wooden house** shifting through **three distinct environments**, acting as a gateway between worlds.

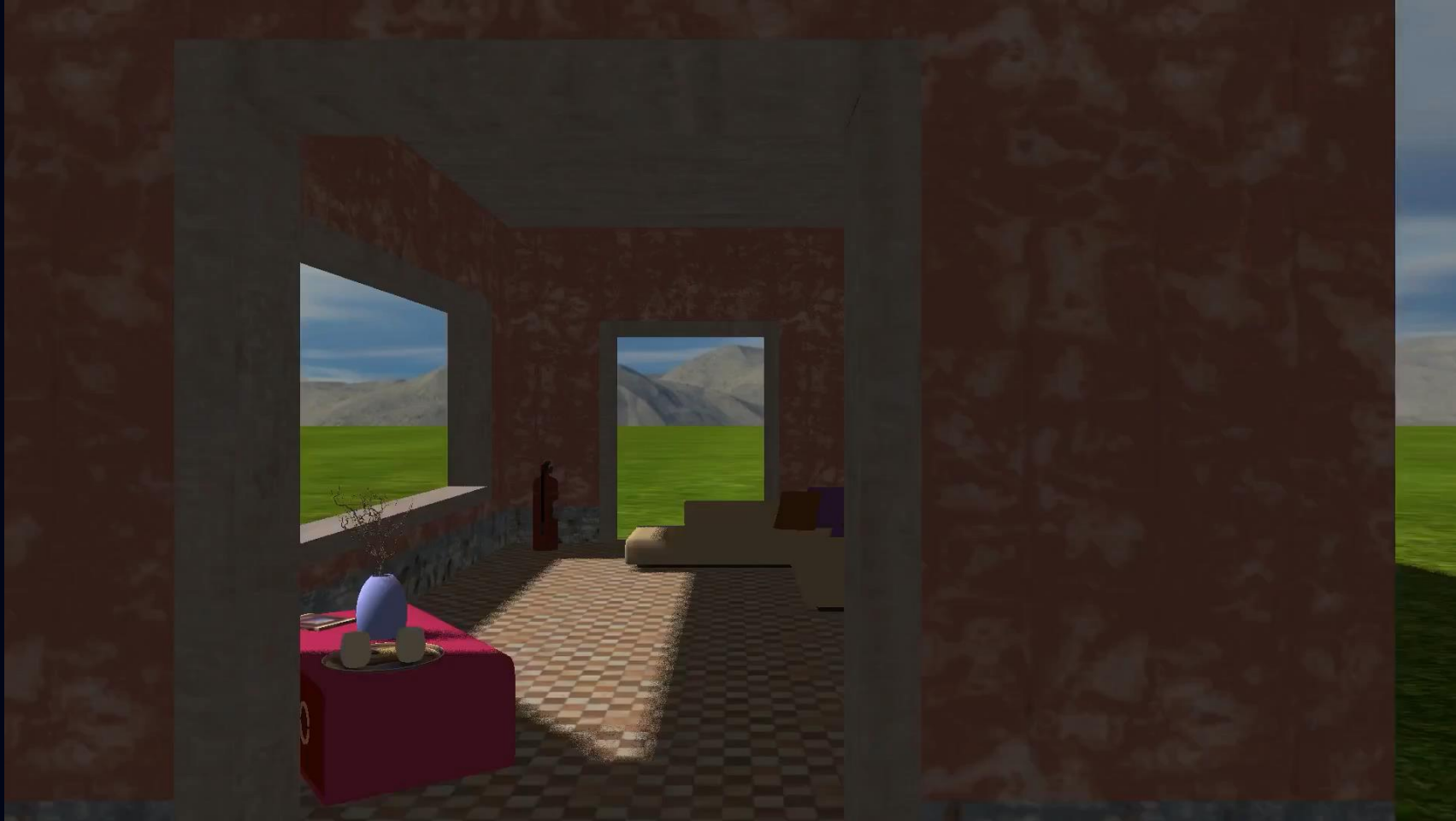
- > Landscapes are viewed through a window.
- > Each scene integrates unique graphic features: technical and artistic.

PLATFORM

OpenGL with C++ and GLSL.



FINAL RESULT



SCENE #0: INTRO



| LIST OF SCENES (w/ Key Features)



VOLCANO

Flame on Objects
w/ **Particle Simulation**



DESERT

Mirage Effect
w/ **Nonlinear Ray Marching**



UNDERWATER

Swimming Fish Swarm
w/ **Skeletal Animation**

SCENE #1: VOLCANO – Sukhun Yang



SCENE #1: VOLCANO TECHNICALS



KEY TECH: Fire Particle Simulation

- > **Particle System:** Effectively managed numerous particles with vectors.
- > **Surface Emission:** For surface fire generation, sampled random vertices & normals from a mesh.
- > **Surface-Guided Particle Ascent:** Vectors that are perpendicular to the surface normal guide particles upward along the mesh surface.
- > **Billboard Rendering:** View matrix components ensure particle quads always face the camera.
- > **Curl Noise:** Utilized simplex-based noise to generate natural, swirling fluid paths.

SCENE #1: VOLCANO ARTISTICS

ARTISTIC FEATURES

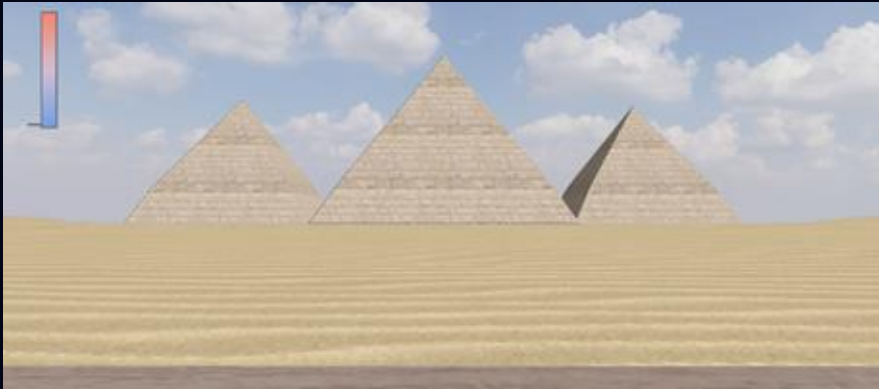
- > **Alpha Blending:** Realism with red/orange particle colors, additive glow, and soft circular alpha blending.
- > **Environment Design:** Volcanic skybox and add assets for a visually rich scene.
- > **Dynamic Elements:** Attached fire trails to meteors and aircraft for a dynamic environment.
- > **Flight Animation:** Curved flight, roll, and Y/Z wave motion to deliver an organic gliding effect
- > **Interactive Controls:** Real-time adjustment of the Dragon's fire spawn rate with keyboard inputs.



SCENE #2: DESERT – Taesun Kim



SCENE #2: DESERT TECHNICALS



KEY TECH: Non-linear Ray Marching

- > **Ray Marching:** To visualize mirage in desert terrain, traced non-linearly refracted light paths by advancing rays at short step sizes with respect to medium states.
- > **Approximated Refraction:** Integrated the Eikonal equation by applying the perpendicular IOR gradient to bend light rays incrementally.
- > **Analytic Intersection:** Detected pyramid/ground hits from local coordinates.

SCENE #2: DESERT ARTISTICS

ARTISTIC FEATURES

- > **Parametric Heat Field:** Maximized mirage realism with a thermal field combining XZ-elliptical heat zones, height decay, and macro/micro temporal noise.
- > **Height-limited Bend Mask:** By confining ray refraction to low Z-depth intervals, the illusion is realistically concentrated above the surface.
- > **Screen-Space Perlin Noise:** Heat haze is highly visible with Perlin noise-based screen distortion via a separate post-processing shader.
- > **Temperature UI:** Visualized current thermal state with a simple UI bar.



| SCENE #3: UNDERWATER – Kanghyeon Cho



SCENE #3: UNDERWATER TECHNICALS



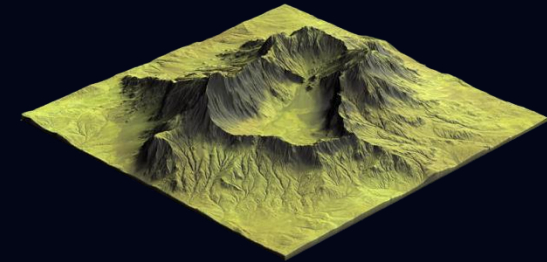
KEY TECH: Underwater Animations

- > **Skeletal Animation:** Implemented swimming motions by loading COLLADA files for small fish and sharks.
- > **Caustics Effect:** Rendered underwater patterns via periodic caustic maps. Projective texture mapping seamlessly apply effects across both terrains and fish models.
- > **Flocking Behavior:** Implemented Boids algorithm (Reynolds, 1986) as the movement policy to simulate natural fish swarming.

SCENE #3: UNDERWATER ARTISTICS

ARTISTIC FEATURES

- > **Distance Fog:** Ambient background color blending to objects based on distance achieved realistic underwater fog attenuation.
- > **Environment Assets:** Enhanced scene richness with various assets including shells, ship, gravel, and repurposed mountain meshes.



Q&A

Thank you for your attention.

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